S.No: 3	Exp. Name: ***Quick sort***	Date:2025-07-28

Aim:

Write a program to perform Quick sort. Display the partial pass-wise sorting done.

Source Code:

quickSort.c

```
// Type Content here...
#include<stdio.h>
void swap(int *a,int *b){
   int temp = *a;
   *a = *b;
   *b = temp;
}
int partition(int arr[], int low, int high){
   int p=arr[high];
   int i=low-1;
   for (int j=low; j<high; j++){
      if(arr[j]<=p){
         i++;
         swap(&arr[i],&arr[j]);
      }
   }
   swap(&arr[i+1], &arr[high]);
   return i+1;
}
void quickSort(int arr[], int low, int high){
   if(low<high){
      int pi=partition(arr, low, high);
      printf("Pass: ");
      for(int i=low; i<=high; i++){
         printf("%d ",arr[i]);
      }
      printf("\n");
      quickSort(arr, low, pi-1);
      quickSort(arr, pi+1, high);
   }
}

int main(){
   int arr[100],n;
   printf("number of elements: ");
   scanf("%d",&n);
   printf("elements: ");
   for(int i=0;i<n;i++){
      scanf("%d",&arr[i]);
   }
   printf("Original array: ");
   for(int i=0;i<n;i++){
      printf("%d ", arr[i]);
   }
   printf("\n");
   quickSort(arr, 0, n-1);
```


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```
    printf("Sorted array: ");
    for(int i=0; i<n; i++){
       printf("%d ",arr[i]);
    }
    printf("\n");
    return 0;
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
number of elements: 4
elements: 5 8 9 4
Original array: 5 8 9 4
Pass: 4 8 9 5
Pass: 5 9 8
Pass: 8 9
Sorted array: 4 5 8 9

Test Case - 2
User Output
number of elements: 6
elements: 5 1 10 8 9 7
Original array: 5 1 10 8 9 7
Pass: 5 1 7 8 9 10
Pass: 1 5
Pass: 8 9 10
Pass: 8 9
Sorted array: 1 5 7 8 9 10